

THE EHRLICHIOSES

Diagnosis and Differential Diagnosis

Enzyme-linked immunosorbent assay (ELISA), dot blot urine immunoassay, rapid immunochromatographic flow assay, and PCR offer the highest sensitivity and specificity for diagnosing ehrlichial infections. However, their use is often limited by lack of availability in endemic regions. Indirect immunofluorescence assay (IFA) and indirect immunoperoxidase assay are highly specific—though somewhat less sensitive—and are more widely accessible, making them the most commonly used diagnostic methods worldwide.

An IFA titer of ≥ 400 is 96% specific but only 48% sensitive. Lowering the cutoff to ≥ 100 increases sensitivity to 78% but reduces specificity to 84%.

The differential diagnosis for ehrlichiosis overlaps significantly with that of endemic (murine) typhus. Common considerations include other rickettsial diseases, leptospirosis, viral hepatitis, and other tick-borne infections such as Lyme disease and babesiosis.

Treatment and Prognosis

Although short courses of doxycycline (e.g., single dose or 3-day regimen) may be effective, a full 14-day course is recommended for optimal treatment. Naturally occurring doxycycline- and chloramphenicol-resistant strains of *Orientia tsutsugamushi* have been identified in northern Thailand. In cases of resistance or intolerance, alternative agents include azithromycin, rifampin, and ciprofloxacin.

With appropriate antibiotic therapy, defervescence is typically abrupt, occurring within 24 hours. In untreated individuals, fever generally persists for approximately 2 weeks.

Prevention

DEET, when applied to skin or impregnated into clothing, is effective in preventing tick and vector attachment, thereby reducing transmission risk. For travelers visiting endemic areas, chemoprophylaxis with weekly doxycycline (200 mg) may provide protection, although this approach is not routinely recommended for general use.

Q FEVER

(Image removed: Q fever illustration)

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Epidemiology

Human ehrlichiosis has been recognized in the United States since 1986. The primary agent of human monocytic ehrlichiosis (HME) is *Ehrlichia chaffeensis*, with white-tailed deer, dogs, and coyotes serving as main reservoirs. *Anaplasma phagocytophilum*, the causative agent of human granulocytic anaplasmosis (HGA), primarily resides in the white-footed mouse (*Peromyscus leucopus*) and white-tailed deer (*Odocoileus virginianus*). The epidemiologic features of HME and HGA are summarized in key reference tables.

HGA is geographically distributed similarly to Lyme disease (*Borrelia burgdorferi*), with endemic areas in the northeastern U.S., upper Midwest, and parts of the Pacific Northwest. The vector for both HGA and Lyme disease is the *Ixodes* tick (e.g., *Ixodes scapularis*, *Ixodes pacificus*). These ticks also transmit *Babesia microti*, *Borrelia afzelii*, and *Borrelia garinii*. Consequently, co-infections with Lyme disease and babesiosis are common in HGA-endemic regions.

The lone star tick (*Amblyomma americanum*) and the American dog tick (*Dermacentor variabilis*) serve as vectors for *E. chaffeensis*. These same ticks can transmit *Rickettsia rickettsii* (causing Rocky Mountain spotted fever) and *Francisella tularensis* (causing tularemia). Therefore, co-infection with these potentially severe pathogens should be considered in any patient with suspected or confirmed ehrlichiosis.

Etiology and Pathogenesis

Ehrlichiae and *Anaplasma* species are small, obligate intracellular bacteria. *Ehrlichia chaffeensis* primarily infects monocytes, whereas *Anaplasma phagocytophilum* targets granulocytes (myeloid cells). These organisms replicate within membrane-bound vacuoles in host cells, forming intracytoplasmic microcolonies known as morulae. Morulae can sometimes be visualized on peripheral blood smears, particularly in granulocytes in HGA cases, and may aid in early diagnosis.